

Compounding the Little Things: Automating the Detection of Incremental Change in Public Spaces, Beginning with Roman Street Shrines

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Abstract

Small, incremental changes in public spaces often go untracked until their cumulative effects become impossible to ignore. This study develops a tool that automates the detection of such changes, beginning with Roman street shrines. Researchers studying whether these shrines deter crime, how sacred imagery compares to secular surveillance like CCTV, and how personal devotion persists in secularizing societies all depend on the same foundational task: knowing which shrines are still active and which have gone quiet. That task is currently performed by hand, with a researcher opening Google Street View, examining each available year of imagery, and manually counting every item on the wall. This study replaces that process with a locally hosted open-source vision language model that analyzes each image and estimates what is present. A working prototype has been tested at a shrine in Rome across imagery spanning 2008 to 2025. The model can produce inconsistent counts between runs and cannot understand the meaning of what it observes, so a human must still verify results. But even reducing a fraction of the manual effort makes large-scale tracking feasible. Because the method works with any fixed location and sequential public imagery, it extends beyond shrines to monitoring infrastructure decay, signage deterioration, and identifying neglected structures in communities where the resources to inspect every street do not exist.

Background

Roman street shrines, known as *madonnelle*, are devotional images mounted on building walls throughout Italian cities, most concentrated in Rome. Muir (1989) traces their origins to ancient Roman *lares*, guardian figures at crossroads, later replaced by images of the Virgin Mary. Unlike typical Madonna depictions, the *madonnelle* stare directly at the viewer. Dear et al. (2019) found through a meta-analysis that images of watching eyes reduce antisocial behavior by 35%, giving behavioral grounding to the long-held belief that these shrines shape public conduct. Whether they still serve that function depends on whether they remain active. Vezzoni and Biolcati-Rinaldi (2015) found that church attendance in Italy has declined since the late 1960s, while Bossi (2023) found that personal faith persists outside formal church structures among younger Italians. Street shrines may be one of the few visible indicators of that persistence, but tracking their activity across hundreds of sites and years of imagery has required manual, frame-by-frame counting in Google Street View, a process too slow to scale.

Recent computer vision research demonstrates that this kind of problem is solvable at scale. Gebru et al. (2017) processed 50 million Street View images to identify vehicles and predict neighborhood demographics, completing in two weeks what would have taken a human 15 years. Naik et al. (2017) used Street View to measure physical urban change over time. Both show that publicly available street-level imagery can reveal patterns invisible to manual observation. Neither has been applied to religious devotion or cultural heritage tracking. There is currently no automated method for monitoring how devotional activity at street shrines changes over time, and without it, the larger questions about sacred surveillance, crime deterrence, and the trajectory of secularization lack the foundational data they require.

Significance

The method this tool uses, retrieving historical street-level images at known coordinates and passing them to a vision model that describes what it sees, is not specific to devotional objects. The same process applies wherever visible change at street level carries meaning. A municipality could use it to prioritize which signs or facades need inspection first. Communities with limited resources could identify and track abandoned or severely decayed structures from imagery that already exists, rather than waiting until consequences become severe. It is often the small, incremental forms of decay, the ones that do not seem urgent on any given day, that compound into conditions people eventually have to live with. The tool runs on consumer-grade equipment, requires no cloud services or institutional access, and uses only public imagery, meaning the responsibility for monitoring does not have to rest with a government or an institution. Within its immediate domain, it provides the baseline data that shrine researchers currently lack. Beyond that, it demonstrates a method that is transferable and operable by anyone with a computer. The tool does not replace human judgment. A vision model is not a stakeholder: it does not understand the consequences of its analysis or the weight of what it counts. A human must still

verify and interpret. But lifting even a fraction of the manual work frees researchers to focus on the questions that require human insight.

Project Design

A working prototype exists. It is a Chrome extension paired with a local Python server. The user navigates to a shrine in Google Street View, clicks the extension, and the server retrieves all available historical panoramas at that location. Each image is sent to a vision language model (Gemma 3, running locally through Ollama) with a structured prompt that asks it to count devotional items in five categories: plaques, flowers, candles, pictures, and other objects. The model returns count in JSON format, and the extension displays a timeline chart with side-by-side comparisons between consecutive years.

Over an 8 - 10-week research period, the first phase (weeks 1 - 2) focuses on accuracy. The model produces useful trend-level data but inconsistent exact counts between runs, which is inherent to how language models generate responses. This phase involves consulting with researchers experienced in vision models to identify improvements to the model, prompt structure, or technical stack. Specific strategies include few-shot prompting (showing the model verified example images before analysis), testing larger models, and refining negative constraints in the prompt to reduce false positives from street clutter. The second phase (weeks 3 - 6) focuses on validation: running the tool across all three sites from the previous manual study, establishing a ground-truth baseline through human counting of a subset of images, and measuring inter-run consistency. The third phase (weeks 7 - 10) tests generalizability on a non-shrine application, finalizes the research paper, and prepares the tool for use by other researchers.

Expected Outcomes and Deliverables

The primary deliverable is a validated, documented tool that researchers can download and run on their own machines to track visible changes at any Street View location over time. The tool is already publicly available at(<https://github.com/ChimaOzonwoye/Kai/tree/main/shrine-change-tracker>). By the end of the research period, the expected additions are accuracy benchmarks comparing model counts against manually verified ground truth, consistency metrics across multiple runs, a completed analysis of all three previously studied shrine sites with direct comparison to the prior manual study, and a research paper documenting the methodology, results, and limitations. A secondary outcome is a preliminary test of the method on a non-shrine application to assess whether the approach generalizes. The expected outcome is not a perfect system but a practical one with clearly documented capabilities and boundaries.

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